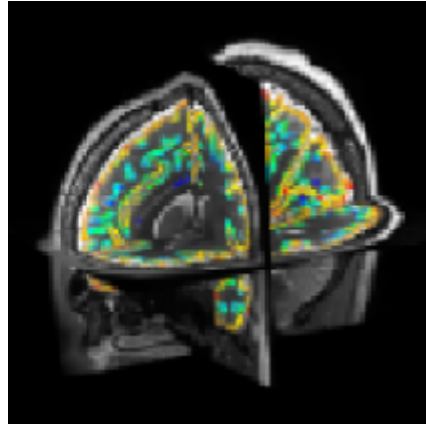
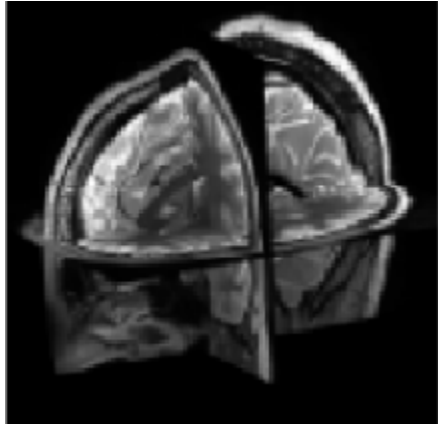


Use Case 4: Mapping cortical structures in brain MRI



You are part of a research team developing an AI model for automatic segmentation of cortical structures in high-resolution brain MRI scans. The goal is to support neuroimaging studies that investigate cortical organization, structural brain changes, and disease-related alterations over time.

The algorithm produces segmentation masks of elongated cortical structures. Each target structure is expected to be represented as a single, non-overlapping anatomical structure, and each voxel should belong to only one label or to the surrounding background.

Researchers are interested in several complementary properties of the segmentation result: the exact location and centerline of the structure, the accuracy of its boundaries, and the estimated volume. Small spatial deviations can matter, especially when comparing subtle anatomical differences across subjects or timepoints.

The reference annotations were created by trained experts. During quality control, the team notices that annotators sometimes disagree on the precise structure boundaries. In addition, a few reference annotations contain small isolated labeling artifacts away from the main structure.

The team does not expect empty segmentations or invalid algorithm outputs under normal operating conditions. However, they want the evaluation to be sensitive to spatially misplaced predictions, especially when prediction errors occur far away from the expected anatomical location.

You now want to validate the segmentation model appropriately and decide which metrics should be used for reporting model performance.

Task description

Study your use case carefully. Then go to: <https://metrics-reloaded.dkfz.de/metric-selection>

Follow the decision process of the Metrics Reloaded toolkit and answer the fingerprint questions based on the information provided in the given scenario. Your goal is to identify the most appropriate performance metrics for the given application.

After finishing the task, save the generated PDF report containing the selected metrics and fingerprint answers. Be prepared to briefly present your use case to the group, including:

- how you interpreted the scenario,
- which fingerprint questions were easy or difficult to answer,
- whether any design choices were unclear or ambiguous,
- and which metrics were ultimately selected by the toolkit.

Please send your PDF report to the lecturers after completing the exercise.